Score-linked practice intelligence from long-form violin training video

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High-level music training produces dense evidence: daily practice videos, repeated repertoire, changing technical constraints, score-free technique exercises, and periodic performance goals. Most of that evidence is difficult to use because it is stored as unstructured media rather than as a score-linked or pattern-linked longitudinal record. We report Curtis, a live AO Labs system that converts messy long-form violin practice videos into structured, evidence-linked practice records using active practice detection, playable audio/video evidence windows, score alignment targets, technique-pattern targets, heat-map targets, repertoire tracking, and passage-level performance observation. The checked live source state at 2026-05-13 12:33 PM EDT used the YouTube channel `@nalalan` as the primary source, indexed 217 public videos, marked 110 as practice candidates, and stored 80 media-sample index entries. The uploaded-video ledger begins at `violin 1`, uploaded on 20 December 2025, and currently counts 46 violin-numbered or date-titled practice logs totaling 213 h 23 min of public session duration; this duration is not counted as active practice time. The daily-record state contains 39 indexed practice days, 3 audio-evidence records, 2 score-audio-only records, 1 detected-note-fragment record, 68 violin-positive sample windows, and 12 withheld non-violin windows. Total practice time is counted only from media windows where violin playing is detected; it is not tied to transcription or score matching. The current measured practice total is 53 min 14 s from 2 h of checked video, which extrapolates to an approximate 94 h 39 min total practice-time estimate across the 213 h 23 min archive until the full low-cost scan completes. All unverified room, setup, laptop, silence, or non-playing samples are withheld rather than counted as measured practice evidence. This PDF is now also the progress tracker for the long-phrase transcription and exact score-matching goal: every meaningful Curtis transcription, score, practice-time, heat-map, repertoire, or observation improvement should update this paper in the same work cycle. The primary interface has three top-level surfaces: a paper card, analyzed practice days, and repertoire. The analyzed-days evidence row reports total

practice time first, then video checked for practice-time detection, then the full uploaded-video archive so the interface does not hide either the multi-hour corpus or the scan coverage. The analyzed-days list keeps practice days chronological and displays exact audio-checked D4 and A4 fragments from the 2026-05-03 Scherzo-Tarantelle practice sample as detected-note evidence, not as score-location matches. Note-match groups are now driven by pitch-class sequence agreement against an available score-derived sequence or a separately labeled reference-audio sequence; the new symbolic-score path parses MusicXML into solo-violin notes and accepts a score match only when the played pitch-class sequence contains at least five contiguous score notes with at least three distinct pitch classes. The displayed score snippet for such a match is rendered from the exact matched symbolic score notes rather than from a guessed scanned-image crop. The Scherzo-Tarantelle target currently uses a 179-note local reference-audio pitch trace, not detected score notes or a parsed symbolic score, so loose reference matches stay hidden from the daily row until score evidence exists. Each daily record withholds embedded playback, paired audio, and practice-time credit until violin playing is detected in the source window, withholds broader sheet-music notation until reliable note extraction supports either score alignment or a score-free technique-pattern record, and withholds exact score-location snippets until a played window has explicit measure agreement rather than only a source-confirmed piece title or broad pitch-sequence match. Single-pitch score crops are withheld unless the highlighted score note has explicit visual note verification. All current Scherzo-Tarantelle single-pitch crops, including the later A4 crop, are rejected after review because the highlighted score note did not match the displayed A pitch. The live system therefore has no accepted single-note score crop; detected D4/A4 audio fragments remain audio-checked transcription evidence only, not score evidence. Draft notation does not render as transcription; repeated single-pitch traces such as a spurious `D D D D` stream remain out of final score-verified notation. The transcription-run PDF now lists all stored detected note series for the day, while note matching uses pitch-class sequence agreement with rhythm and octave ignored except for visually verified octave-specific score anchors. Curtis keeps source-confirmed pieces, explicit calibration titles such as scales and arpeggios, official YouTube Data API public-reference metadata, nearest reference targets, and score context available without turning untrusted note events into accepted score evidence. Manual day-level source labels can be entered from the analyzed-day record, but these labels only name the practice source; they do not unlock score snippets, heat maps, readiness scores, or repertoire progress without matching audio and score evidence. Public labeled YouTube

references seed likely reference targets but do not become repertoire, audio fingerprints, or judged performance evidence until a permitted media path and score or pattern alignment exists. Active material with no confirmed score is labeled as piece-or-exercise pending rather than forced into a wrong repertoire guess. Repertoire entries are promoted only from confirmed daily-record evidence and keep possible labels, user-rejected false labels, weak evidence, and readiness scores inferred only from piece identification separate from confirmed repertoire and judged playing quality. The system uses GPT-5.5 for vision and text review and separate audio models for identification and verification, while preserving explicit limits: Curtis admission cannot be predicted from current samples, weak audio/video evidence is marked unjudged, whole-session transcription is not yet complete, and the system reports practice signals rather than admissions probability. The contribution is an experimental method for making practice visible, searchable, measurable, and tied to actual musical evidence.

1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (*1*, *2*). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis addresses this problem as a live software system rather than as a portfolio page (*3*). The system indexes public practice media, separates uploaded video duration from detected violin-playing practice time, records technical observations, and keeps evidence tied to clips and generated notation. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and identify the specific passage-level blockers that can be tested in subsequent recordings.

The core pipeline is video → active practice detection → clips → music transcription → repeat grouping → score matching → heat map → repertoire update → passage-level observations. The output is intentionally small: analyzed practice days and an automatically updated repertoire list, both grounded in clips, notation, active playing time, and confidence limits.

The system is also part of AO Labs' broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (*4*). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

2 System state

The production state uses the live backend at `https://curtis.aolabs.io/api/curtis/media-status`. The checked source state at 2026-05-13 12:33 PM EDT reported the values in Table 1. These are operating facts, not outcome claims.

Table 1. Curtis source state checked at 2026-05-13 12:33 PM EDT.

Signal	Current value
Primary source	YouTube channel `@nalalan`
Inventory	217 indexed videos
Practice candidates	110 videos
Long-form candidates	90 videos
Uploaded-video marker	`violin 1`, uploaded 20 December 2025
Uploaded-video ledger	46 videos, 213 h 23 min uploaded session duration
Estimated total practice time	94 h 39 min extrapolated from the checked-window active-playing ratio; approximate until the full archive scan completes
Measured practice time	53 min 14 s of detected violin-playing footage; this is independent from transcription
Video checked for practice time	2 h of the 213 h 23 min uploaded-video ledger
Unchecked archive duration	211 h 23 min still awaiting violin-playing detection
Media samples	80 indexed windows; 68 violin-positive; 12 withheld
Reviewed sections	80 sections
Curtis-focused findings	80 findings
Latest dated practice log	5-6-26
Review model	GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification
Confirmed labels	5/1 Haydn Symphony No. 94 IV; 5/2 and 5/3 Wieniawski Scherzo-Tarantelle
Training sources	3 source-confirmed practice sources
Score-reference targets	3 configured
Reference training	3 source-confirmed score targets, 22 pitch/rhythm windows, title-labeled calibration sources for future scale/arpeggio uploads, and 4 official YouTube Data API public-reference queries kept separate from repertoire claims and audio evidence
Analyzed practice days	39 indexed days; 3 practice-time measured daily records; 3 audio-evidence records; 2 score-audio-only records; 1 detected-note-fragment record; 3 confirmed source labels with score targets and 0 exact score-matched snippets
Machine pitch events	Detected-active-section onset-aware pYIN plus onset-bounded spectral pitch rescue for fast passages, audio-agreement counts, pitch-sanity filtering, repeated-pitch-collapse guarding, practice-coverage fields, title-labeled calibration handling, and fingerprints stored only when quality gates pass; broad candidate micro-transcription is withheld, while detected note series are preserved in the transcription PDF and promoted to note-match groups only after pitch-class sequence agreement with an available score-derived or reference-audio sequence
Specific observations	Rendered only when grounded in reliable score-linked evidence or repeated score-free technique-pattern evidence
Note-match windows	15 pitch-sequence note-match groups; 0 exact score-aligned windows
Daily evidence surface	Local sample playback only for violin-positive windows, notation withheld until reliable note/rhythm verification, score snippets hidden until played-to-score agreement, and explicit whole-session transcription limit
Curtis-level completion	Not scored when the available evidence only identifies repertoire
Current focus	Capture a playable audition-view take
Current constraint	Record 30 seconds of sustained tone
Boundary	Curtis admission cannot be predicted from current samples

2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal records because the title contains a practice date and the duration preserves session-level evidence. The uploaded-video ledger starts at the `violin 1` marker and counts only violin-numbered or date-titled practice logs, so unrelated long public videos are retained in inventory without being silently counted as violin practice. This ledger is not the active-practice total. Active practice is measured only after media processing identifies windows where violin playing is actually occurring.

2.2 Review evidence

The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

3 First readout

The first live readout is experimental and intentionally conservative. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples, stored section reviews, and title-confirmed practice records. The home screen has three top-level surfaces: a paper card, analyzed practice days, and repertoire. The first evidence row now reports total practice time first, then the video already checked for violin-playing, then the 213 h 23 min YouTube practice archive; this prevents the public practice ledger from being confused with either transcription coverage or uploaded-video duration. Each analyzed day groups same-day videos, preserves uploaded duration, and names confirmed or uncertain piece evidence without promoting guesses. Playable local media samples, paired audio/video evidence windows, and practice-time credit are now withheld unless violin playing is detected in the source window. The owner-sync sampler now reuses locally cached captured media, uploads cached violin-positive windows before blind new captures, expands around already-found violin-positive windows, and can switch into full-check mode to walk every 90-second window in order instead of staying limited to sampled windows; the live analyzed-days surface currently exposes violin-playing practice-time evidence for 5/1, 5/2, and 5/3. Broad sheet-music notation is also withheld until reliable note/rhythm verification, so audio evidence cannot appear as a failed transcription. The current visible notation is limited to exact audio-checked D4 and A4 fragments from 5/3, paired with generated local clips for the same source sample windows. Actual score images are hidden until exact score-location agreement exists; no single-note Scherzo-Tarantelle score crop is currently accepted. A source-confirmed piece title creates a score target, and a detected pitch-class sequence match can show the played clip and transcription, but broad score-region estimates do not render as sheet-music snippets. The daily record includes a compact source-label field so a missing or wrong day title can be corrected without letting that correction masquerade as score alignment. The analyzed-days surface no longer promotes broad candidate transcription while candidate micro-transcription fails audible match review; those windows remain hidden support data or audio evidence only. Active scans with no confirmed piece are treated as possible repertoire or score-free technique exercise evidence rather than as failed score matching. The repertoire surface is evidence-backed: an entry appears only when a daily record carries confirmed source evidence, and each entry keeps practice dates, clips, score targets, practice time when measured, and current blocker limits. Generic etude/caprice evidence remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable against the source window. These statements are not generic pedagogy; they are extracted from stored review findings, sample indexes, source corrections, public-domain score targets, pitch-quality checks, and the uploaded-video ledger.

Table 2. Current evidence classes and limits.

Class	Stored signal	Limit
Audio	Tone, intonation, shifts, time, articulation	Some audio reviews fail or cannot inspect the excerpt cleanly
Video	Setup, posture, bow lane, visible hand/bow evidence	Still frames can hide motion, sound, and transition accuracy
Metadata	Dates, duration, channel, view count, practice-candidate labels	Metadata cannot judge playing quality by itself
Progress plan	One focus, practice constraint, short session plan	The plan is a next experiment, not an outcome prediction

Current transcription limit. Machine note output now has three display gates. First, the media window must be classified as violin-positive before it can appear as playable evidence or count toward practice time. Second, the note stream must audibly match the paired audio before any notation can render. Third, repertoire passages need score alignment, while score-free exercises need repeated-pattern grouping before they can become full practice transcription. Source-confirmed score targets are also display-gated: broad score snippets are hidden unless the corresponding played window has explicit score-location agreement, and single-pitch score crops require explicit visual note verification before they can render. Score-note sequences are now source-gated before matching: a scanned PDF plus guessed `scorePitchClassSequences` is ignored unless the note sequence has an explicit symbolic-score source, manual source confirmation, MusicXML/source-confirmed status, or exact score-location metadata. Curtis now includes a symbolic-score parser for inline MusicXML, local `.musicxml`, and local `.mxl` targets. That path extracts score notes, measures, pitch classes, octave-aware note names, and duration fields; a visible score match requires at least five contiguous played pitch classes to appear in the parsed score and at least three distinct pitch classes in the match. One-note overlaps and repeated single-pitch runs are therefore rejected as score evidence. When the symbolic gate passes, the score panel is generated from the matched score notes and measure label, so a displayed score-side A must come from an actual parsed A rather than from a visually guessed crop. All current Scherzo-Tarantelle single-pitch crops are rejected and withheld because visual note review found that the highlighted score notes did not match the displayed A pitch. Unverified room, setup, laptop, silence, or non-playing samples remain withheld, and unverified pitch traces remain hidden support data rather than visible notation. A spectral-onset rescue path estimates pitch inside short onset-bounded violin segments when the pYIN path collapses, while the repeated-note trace guard prevents fast or complex playing from becoming a dominant repeated pitch, records the quality mode, and keeps unusable traces out of accepted score evidence. The current micro-transcription path stores candidate short contiguous runs, but it does not render them as sheet music or count them as solved transcription until the displayed notes pass paired-audio listening review. The transcription-run PDF preserves all stored detected note series for the day. Note matching uses pitch-class sequence agreement, ignores rhythm and octave for the initial grouping step, and can promote detected Scherzo-Tarantelle fragments only when the played pitch-class sequence appears in the local reference-audio pitch trace or another available score-derived/reference sequence; this is not treated as detected score notation unless exact score-location metadata or parsed symbolic score notes exist. A separate audio-checked fragment gate can render a note when pYIN and YIN agree on the same semitone over a short stable window; the present live examples are D4 and A4, not a full passage and not a score-location match. Full-session transcription remains incomplete until every playing section in the long-form practice videos is extracted, verified as violin-positive, verified against the audio, and aligned either to a score section or to a repeated technique pattern.

4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. The sampler now treats violin presence as a hard evidence boundary: playable clips and practice-time credit appear only after violin playing is detected in a media window, while notation waits for reliable note/rhythm verification and either score alignment or score-free pattern grouping. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings easier to compare against earlier ones.

The main open problem is repertoire identification, score-free exercise representation, and evidence quality. YouTube metadata can identify candidate videos and public labeled reference sources, and the current owner-sync path can provide direct audio/video samples, but unclear camera framing, missing score/title context, improvised technical patterns, and short excerpts still prevent reliable piece naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. The current training ledger separates source-confirmed labels from title-labeled calibration uploads, public YouTube reference metadata, independent audio-title matches, score-free or piece-pending active windows, failed unverified machine pitch/rhythm windows, failed pitch-collapse windows, withheld candidate micro-transcription windows, audio-checked single-note fragments, detected note series, pitch-class reference matches, and true score-aligned windows. It also stores reference targets for the three confirmed source labels across two works so later passes can ask for section-level matching instead of repeating broad title guesses. The sampler classifies candidate windows using active audio, violin-range voiced pitch, onset density, pitch-class diversity, spectral features, and minimum practice-duration gates before a sample can become visible evidence. The transcription path converts sampled media to mono audio, finds audio-active playing sections, runs pitch tracking only on those sections, normalizes and harmonic-isolates each section for pitch tracking, estimates monophonic violin pitch with an onset-aware pYIN pitch track bounded to the violin range, detects onset boundaries so repeated attacks on the same pitch are not silently merged into one sustained note, runs a second spectral pitch estimate inside short onset-bounded segments when fast playing defeats pYIN, compares pitch diversity across candidate note streams, checks whether selected note events agree with an independent spectral or pYIN/onset detector, rejects out-of-range pitch artifacts, requires sustained pitch changes for pitch-track segmentation, removes very short low-confidence pitch blips, marks isolated octave-flip corrections as uncertain, rejects repeated-pitch collapse, estimates tempo, stores detected note series for audit/PDF review, and builds pitch-class, interval, rhythm, and contour fingerprints only when quality gates pass. Daily records now carry transcription quality, material status, and practice-coverage states, and the primary interface withholds media playback unless the source window is violin-positive, withholds broad machine notation until note/rhythm verification, displays only audio-checked fragments when the stricter gate passes, hides loose reference-audio matches from the daily row, shows a source-score crop only when the highlighted score note has explicit visual note verification, and withholds broader score snippets until exact score-location alignment exists. A stronger system should align repertoire events against a symbolic or optical score representation, group score-free technical patterns by repeated note/rhythm fingerprints, identify movement sections or measure ranges when a score exists, compare against reference recordings available through permitted media, and then judge articulation, timing, tone, and tempo against the aligned passage or repeated pattern.

The broader experiment is whether a practice system can make musical development legible over time. A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, endurance, and audition delivery are improving across dated recordings. It should also preserve the source evidence behind each claim so that the dashboard does not become motivational text detached from the actual playing.

5 Roadmap to solved long-phrase transcription

This section is the working plan for making Curtis able to produce long, trustworthy audio–video–transcription–score evidence from Alan's actual violin practice. It is intentionally part of the paper rather than a separate planning note. Whenever Curtis makes real progress toward full-session active-practice measurement, accurate transcription, score matching, heat maps, repertoire updates, or Curtis-level observation, this PDF should be updated with the new state, the acceptance evidence, and the remaining blocker.

5.1 Current tracker state

The tracker starts from the 2026-05-13 12:33 PM EDT live state. The system has a real media ledger and a conservative evidence gate, but long-phrase transcription is not solved. The existing trusted musical output is limited to short audio-checked note fragments. The system has no accepted exact score-aligned Scherzo-Tarantelle window, and the previously displayed single-note score crops are rejected because the highlighted score-side note did not match the displayed note label.

5.2 Definition of solved

Long-phrase transcription is solved for Curtis only when the system can take Alan's long-form practice archive and produce evidence groups that a musician can trust without manually checking every note. The complete goal is not merely a model that prints plausible notation. The solved system must satisfy all of the following conditions.

1. **Full archive accounting.** Every practice video from the strict practice ledger is assigned to the correct practice day, has exact uploaded duration, and has a measured or explicitly pending active-practice state.

2. **Active violin time.** Total practice time means intervals where Alan is actually playing violin. Talking, tuning, setup, walking, camera adjustment, silence, scrolling, and other non-playing intervals are excluded.

3. **Local evidence playback.** Every accepted transcription snippet has a playable local video clip and audio clip in the Curtis site. Accepted evidence does not rely on a YouTube embed.

4. **Accurate note events.** The transcribed notes match the paired audio, including repeated notes, fast scalar runs, arpeggios, shifts, string crossings, rests, and obvious restarts. A fast passage must not collapse into a false repeated pitch stream such as `D D D D`.

5. **Rhythm and timing.** The system estimates onsets, durations, rests, tempo changes, slow-practice stretches, restarts, and repeated attempts. When rhythm is uncertain, the notation must mark uncertainty rather than invent a clean answer.

Table 3. Progress tracker baseline for the long-phrase transcription goal.

Tracker item	Baseline state
Goal	Produce compact daily evidence groups: original score snippet, generated transcription, local audio, local video, heat-map context, and specific observation.
Current accepted transcription	Short audio-checked D4/A4 fragments only; not a full phrase and not a score-location match.
Current accepted score match	0 exact score-aligned snippets.
Current archive coverage	213 h 23 min uploaded practice video across 46 strict-ledger videos and 39 practice days; 2 h checked; 53 min 14 s measured active violin; 211 h 23 min still unchecked.
Current estimate	94 h 39 min total active practice estimated from checked windows; not final.
Current user burden to remove	Alan is still forced to inspect whether a note is real, whether a score box points to the correct pitch, whether audio matches notation, and whether practice time means playing rather than uploaded duration.
First tracker update	This PDF now defines the full implementation plan and becomes the required progress record for future Curtis work.
2026-05-13 11:57 AM EDT code update	Backend coverage ledger, evidence-correction route, and benchmark-fixture scaffold added. Checked video time now includes sampled non-playing windows; total practice time remains measured active playing only. Live deployed coverage now reports 213 h 23 min uploaded video, 2 h checked video, 53 min 14 s measured active practice, and 211 h 23 min unchecked. Wrong score-note corrections can be stored as rejected benchmark cases. Full archive scan and long-phrase transcription remain unsolved.
2026-05-13 12:33 PM EDT code update	Low-cost active-practice scan route implemented. The new scan persists active violin-playing sub-intervals from stored violin-positive media samples, keeps checked non-playing samples separate from practice time, exposes `/api/curtis/active-practice-scan`, adds an active-scan run endpoint, queues remaining strict-ledger windows for owner media capture, and feeds persisted active intervals into daily records and the coverage ledger. Local unit tests verify active sub-interval extraction, negative-sample withholding, and coverage accounting. Live scan execution is pending deployment of this endpoint.

6. **Professional notation.** Displayed notation uses a real notation engine or a notation-quality rendering path. Treble clef placement, staff scaling, note placement, stems, beams, rests, spacing, key signature, and measure layout must look like real sheet music, not a hand-drawn mockup.

7. **Score-source truth.** A score snippet can appear only when the score-side notes are source-confirmed. Scanned images without a verified note map are not enough. A displayed A must point to an actual A in the score.

8. **Score alignment.** For known repertoire, the played note sequence must align to a specific location in the score. The match can ignore rhythm during the first pitch-class search, but accepted score display needs exact score-note identity and a stable location.

9. **Score-free exercise handling.** If Alan invents an exercise or practices a technique pattern that is not in the score, Curtis should still transcribe and group the pattern. It should not force the pattern into Scherzo-Tarantelle, Haydn, or another piece.

10. **Repeat grouping.** Repeated attempts of the same passage are grouped as attempts, loops, restarts, or slow/faster passes. The UI should not dump the same measures repeatedly.

11. **Heat maps.** Practice density, repetition density, problem density, and improvement layers appear on actual score coordinates when a score match exists, or on the generated pattern/transcription when the material is score-free.

12. **Curtis-level observations.** Observations are specific and evidence-tied: what happened, where it happened, how often it happened, whether it improved, and why the issue blocks Curtis-level readiness. Generic advice is not accepted.

13. **Compact interface.** Expanded daily entries stay compact. The useful unit is the grouped score–transcription–audio–video pair, not a large dashboard or a long explanatory panel.

14. **Regression protection.** Wrong score-note boxes, failed transcriptions, unmatched notation, stale piece labels, and non-playing practice-time credit cannot silently reappear after a later change.

5.3 Acceptance gates

The following gates define whether progress is real. If a change fails one of these gates, it is not counted as solved, even if the UI looks better.

Archive gate.

The strict practice ledger must list every video from `violin 1` forward that is either violin-numbered or date-titled as practice. Same-day videos must merge into one daily record. Each record stores source video id, title, uploaded date, inferred practice date, uploaded duration, and processing coverage.

Practice-time gate.

Active violin time is the sum of detected playing intervals, not the uploaded duration. The API must expose measured active time, checked video time, unmeasured video time, and estimate status separately. The UI label should use total practice time for active violin time and uploaded video for total source duration.

Evidence-media gate.

A transcription line is accepted only if the exact audio and video window are available locally, playable in the same Curtis page, and time-linked to the source record.

Note gate.

A displayed note must agree with the paired audio. The note label, octave where claimed, and staff position must be internally consistent. If the audio says A4, the rendered staff note must be A4, not F, G, B, or a visually adjacent note.

Score gate.

A score snippet must come from a verified symbolic score, a manually verified note map, or a rendered score source where the highlighted note identity is known. Scanned-image crops may be used only after note-coordinate verification.

Single-note gate.

A one-note score display is allowed only as a calibration anchor, and only if both sides explicitly show the same pitch name and the score-side coordinate is verified. It must not be used as proof of a long passage.

Phrase gate.

A real phrase match requires a contiguous pitch-class sequence in the score or pattern source. Rhythm can be ignored for early matching, but accepted phrase display must keep the exact played sequence, the exact score sequence, and any omitted or uncertain notes.

Notation gate.

Hand-built notation SVG is not the long-term display path. The production path must use a professional renderer or renderer-equivalent layout with verified clef, key, staff, note, rest, spacing, and responsive behavior.

Exercise gate.

If no score match exists and the material forms a repeated exercise, the system must label it as an exercise or pattern, preserve the generated transcription, and avoid adding a fake repertoire entry.

Observation gate.

A Curtis-level observation must cite the clip, transcription event, score measure or pattern segment, issue type, repeat count, direction of change, and readiness consequence.

UI gate.

The day row stays chronological, the expanded state is compact, and only grouped evidence appears by default. Raw untrusted transcription stays hidden or goes to the transcription-run PDF with clear status.

Paper gate.

Any meaningful progress changes this section, the system-state table, and the limitation text in the same work cycle.

5.4 Required source material

The implementation cannot be completed by code alone. It needs a source bundle that makes the audio, score, and corrections checkable.

1. **Raw or locally captured practice media.** Curtis needs stable local access to audio/video windows from the YouTube archive or owner-provided files. Metadata alone cannot produce practice time, clips, or transcription.

2. **Source-confirmed piece names by day.** Alan can supply piece names when automatic detection is uncertain. These labels should seed matching but never become score evidence by themselves.

3. **Symbolic scores for known repertoire.** For each piece, Curtis needs MusicXML, MEI, a verified MuseScore export, or an equivalent symbolic note map with measure numbers, pitches, durations, voices, and staff positions.

4. **Score images tied to symbolic coordinates.** To show actual score snippets, the image/PDF rendering must be linked to the same symbolic note ids or verified coordinate boxes.
5. **Manual corrections.** Alan or a reviewer must be able to correct a note, octave, onset, rhythm, score location, or false match. These corrections become training and regression data.
6. **Calibration recordings.** Labeled scales, arpeggios, open strings, repeated notes, shifts, slow exercises, and fast exercises help tune pitch detection before attempting long phrases.
7. **Gold benchmark clips.** A small set of manually checked audio–transcription–score pairs is required before the app can claim improvement. The benchmark must include easy anchors, repeated notes, fast runs, arpeggios, and real Scherzo-Tarantelle fragments.

5.5 Implementation plan

Phase 0: Make the paper the tracker

Status: complete in this revision. The first required step is to make the plan durable in the Curtis paper PDF. The paper now records the definition of solved, the current baseline, the gates, the phase plan, and the update protocol. Future changes are not complete unless the paper is checked or updated.

Phase 1: Build the canonical evidence ledger

The backend needs a stricter data model for practice evidence. The current modules already include the daily-record builder, media sampler, transcription engine, symbolic-score parser, score-asset layer, reference corpus, and correction layer. The next version should make the following entities explicit:

- `source_video`: immutable source id, title, URL, uploaded time, duration, practice-date basis, and ledger inclusion basis.
- `practice_day`: date, list of source videos, uploaded duration, active-practice coverage, piece labels, and status.
- `active_interval`: source id, start, end, confidence, detector version, reason for inclusion, and reason for exclusion when withheld.
- `media_clip`: local video path, local audio path, waveform path, source interval, duration, and playable route.

- `note_event`: onset, offset, pitch class, octave, cents deviation, confidence, detector sources, uncertainty reason, and correction state.
- `score_note`: piece id, movement, measure, voice, pitch class, octave, duration, note id, and rendered coordinates.
- `match_candidate`: played note ids, score note ids or exercise-pattern ids, match score, mismatch list, and display eligibility.
- `accepted_evidence`: only the groups that pass the gates; these are the only groups the main UI can show.
- `rejected_evidence`: wrong notes, wrong score boxes, false labels, failed transcriptions, and model guesses that must not reappear.

The acceptance condition for this phase is that a wrong A-to-B or A-to-G score crop can be stored as rejected evidence and cannot be selected by the display builder again.

Phase 2: Complete the full active-practice scan

The practice-time problem is separate from transcription. Curtis should first scan the full 213 h 23 min uploaded practice ledger for violin-playing intervals with a low-cost pass. The pass should use audio energy, violin-range voiced pitch, onset density, spectral shape, and visual fallback only where necessary. It should not require full transcription.

The fast pass should walk every practice video in chronological order and store approximate active intervals. A second exact pass should refine boundaries around detected regions. The UI should show three numbers for every day: uploaded video time, checked video time, and total practice time. While scanning is partial, the estimate stays labeled as estimated. When every video has been scanned, the estimate should be replaced by measured total practice time.

Acceptance for this phase:

- every strict-ledger video has a coverage record;
- every day has uploaded duration and active-practice status;
- non-playing windows are excluded;
- active-practice total can be recomputed from stored intervals;

- the paper table is updated with checked duration, active duration, unmeasured duration, and estimate status.

Phase 3: Generate stable local evidence clips

Every accepted interval and every candidate transcription window needs local media. Curtis should generate short MP4 clips and audio clips with stable IDs derived from source video id and time range. The browser should play the local clip and audio without YouTube embedding. The audio used for transcription must be the same audio the user can play.

Acceptance for this phase:

- each accepted transcription event links to local video and local audio;
- the site can play the clip in the same window;
- source time, local file, and rendered transcription agree;
- dead local media routes fail the evidence gate rather than rendering a broken accepted group.

Phase 4: Build the verified score library

The current weak link is score-side truth. Curtis must stop relying on guessed scanned-score boxes. For Wieniawski Scherzo-Tarantelle and any future piece, the score library needs symbolic notes and rendered coordinates.

The best path is:

1. acquire a legal public-domain score image or PDF;
2. obtain or create a symbolic MusicXML/MEI/MuseScore representation;
3. render the symbolic score to image or SVG with stable note ids;
4. map each rendered note to pitch, octave, measure, and coordinates;
5. manually verify a representative set of anchors, especially A, D, G, B, fast runs, ledger lines, accidentals, and octave-displaced passages;
6. reject any score source where the note map cannot distinguish adjacent notes reliably.

Acceptance for this phase:

- a score-side A in the UI is generated from an actual score-side A;
- the rendered crop includes enough context to be legible;
- the crop does not cut off clefs, notes, beams, stems, or measure context;
- single-note anchors are visually verified before display;
- phrase snippets use the exact matched symbolic notes, not a guessed image region.

Phase 5: Replace fragile note extraction with a multi-stage transcription engine

The note detector needs to work on violin practice, not only on clean isolated notes. The engine should process active intervals in stages:

1. normalize audio and remove long silence;
2. isolate violin-dominant energy where possible;
3. detect onsets and repeated attacks before pitch smoothing;
4. run multiple pitch estimators over the same window;
5. separate stable pitch, glissando/shift, vibrato, noise, and uncertain frames;
6. group frames into note events by onset, pitch stability, and duration;
7. preserve fast pitch changes instead of smoothing them into one repeated pitch;
8. estimate rests and pauses between attempts;
9. detect double stops or chords as multi-note events when monophonic assumptions fail;
10. attach confidence and uncertainty to every event;
11. store rejected alternatives so repeated failures can be debugged.

Acceptance for this phase:

- open strings and simple scales transcribe correctly;

- repeated notes remain repeated notes with distinct onsets;
- arpeggios become changing pitch sequences, not a repeated-note collapse;
- obvious fast runs preserve the contour and most pitch classes;
- uncertain events are labeled uncertain rather than shown as clean notation;
- every displayed note can be played from its paired audio.

Phase 6: Make transcription score-aware

For known repertoire, Curtis should not transcribe in isolation and then guess the piece later. The user-supplied or detected piece name should provide a score prior. The aligner should search the score for pitch-class sequences that match the audio-derived note stream, allowing restarts, skipped notes, slow practice, repeated loops, tempo changes, and rhythm mismatch during early search.

The alignment should use multiple layers:

- pitch-class subsequence search for easy anchors;
- interval-contour search when absolute pitch is uncertain;
- dynamic time warping for variable tempo;
- Viterbi-style path selection through the symbolic score for longer passages;
- restart and loop detection when the path jumps backward;
- score-free fallback when no score path is plausible.

Acceptance for this phase:

- one-note anchors pass only when the score coordinate is verified;
- three-to-five-note sequences can be matched to exact score notes;
- longer phrases show played notes beside matched score notes;
- mismatches are visible rather than hidden;
- rhythm mismatch can be tolerated during matching but not during final notation labeling;
- accepted score matches survive a musician spot-check.

Phase 7: Support score-free technique exercises

Alan may invent an exercise during practice. Curtis must treat this as first-class evidence. If the note stream is coherent but does not match the named score, the system should cluster repeated patterns, render generated notation, and label the material as a technique pattern or exercise.

Acceptance for this phase:

- repeated made-up patterns group together;
- the generated transcription is linked to audio/video;
- no fake score snippet appears;
- the repertoire entry records technique-pattern evidence separately from piece evidence;
- heat maps can show pattern density without pretending there is a score location.

Phase 8: Add manual correction and training

Fully automatic transcription will not become reliable without corrections. Curtis needs a correction UI that allows Alan to say: this note is A, this staff note is wrong, this score box points to B, this clip is non-playing, this rhythm is wrong, this is Scherzo-Tarantelle, this is a made-up exercise, or this score match is false.

Every correction should become data:

- corrected note events become gold labels;
- rejected matches become negative examples;
- accepted matches become benchmark fixtures;
- corrected piece names update source labels without unlocking score snippets by themselves;
- corrections invalidate stale derived evidence.

Acceptance for this phase:

- Alan can fix a wrong note or score box without editing code;
- the corrected example appears in the benchmark set;

- the same failure cannot reappear after rerendering;
- the PDF tracker records the correction count and benchmark count.

Phase 9: Build the benchmark suite

Curtis needs objective tests before claiming improvement. The benchmark should include:

- open-string clips;
- G, D, A, and E major/minor scales where available;
- arpeggios;
- repeated single-note attacks;
- fast scalar runs;
- shifts with sliding or unstable pitch;
- Scherzo-Tarantelle fragments with verified score locations;
- Haydn fragments with verified score locations;
- score-free exercises;
- non-playing windows that must not count as practice time.

Metrics should include note precision, note recall, pitch-class accuracy, octave accuracy when claimed, onset error, duration error, false-note rate, repeated-note recall, phrase-match precision, phrase-match recall, score-coordinate accuracy, false accepted score match count, active-practice precision, active-practice recall, and UI evidence completeness.

Acceptance for this phase:

- tests fail on the known A-to-wrong-score-note examples;
- tests fail on repeated-pitch collapse;
- tests fail when accepted notation lacks paired audio/video;
- tests fail when practice time includes non-playing footage;
- tests pass only when the displayed evidence is actually trustworthy.

Phase 10: Use professional notation rendering

The current custom notation path is only a temporary evidence display. The solved system should render notation from MusicXML, MEI, or another structured notation representation through a professional renderer or renderer-equivalent layout. The renderer must handle treble clef placement, staff scale, ledger lines, rests, stems, beams, ties, slurs where known, key signatures, accidentals, note spacing, measure widths, and responsive crop behavior.

Acceptance for this phase:

- treble clef is correctly centered around the G-line spiral;
- the staff note for A4 appears in the A4 position;
- key signature and accidentals follow the chosen notation policy;
- snippets look publication-quality at desktop and mobile widths;
- the same notation source can export to the transcription-run PDF.

Phase 11: Build heat maps from matched coordinates

Heat maps should be computed from actual practice behavior. For score-matched passages, each matched score note or measure receives density values. For score-free exercises, each pattern segment receives density values.

The layers are:

- practice density: active time spent on the passage;
- repetition density: number of attempts or loops;
- problem density: restarts, stops, slowdowns, unstable notes, and failed attempts;
- improvement: later attempts cleaner than earlier attempts.

Acceptance for this phase:

- a heat-map mark can be traced to clips and note events;
- a score-free exercise can still have a pattern heat map;
- heat maps do not appear as decorative overlays when no match exists;

- the day-level heat map and repertoire heat map use the same evidence.

Phase 12: Generate specific Curtis-level observations

Observation generation should happen after transcription and alignment, not before. The observation engine should compare attempts of the same passage and classify issues as pitch, rhythm, bow, shift, tone, articulation, tempo, consistency, musical shape, or visible tension.

The output for each main passage should include:

- likely passage or exercise pattern;
- evidence clip;
- transcription snippet;
- score snippet when verified;
- exact issue;
- number of attempts where it occurred;
- whether it improved, stayed the same, or got worse;
- why it blocks Curtis-level readiness.

Acceptance for this phase:

- observations cite evidence;
- generic advice is absent;
- a day ends with one specific main blocker;
- rejected or uncertain transcription does not feed readiness claims.

Phase 13: Keep the UI compact

The main page should show the paper card, analyzed practice days, and repertoire. Expanded daily entries should show one or a small number of grouped evidence pairs, not a full debug dashboard. The day marker should remain visible on the left so boundaries between days are clear. Videos should be small enough that a day entry fits on screen when possible.

Acceptance for this phase:

- one daily group contains score, clip, audio, transcription, and the minimum useful status;
- raw text explanations are removed from the main view;
- chronological order is stable;
- desktop and mobile render without overlap, excessive height, or broken media controls;
- unresolved material is visible as state, not as a fake dashboard.

Phase 14: Automate paper updates

The paper is now the progress tracker. Future meaningful Curtis changes should update:

- live state numbers;
- solved gates;
- benchmark metrics;
- accepted score-match count;
- accepted transcription duration;
- full archive scan coverage;
- remaining blockers;
- limitations.

Acceptance for this phase is procedural: after every Curtis change that affects transcription, score matching, practice time, clips, heat maps, repertoire, observations, or evidence gates, the LaTeX paper is checked, rebuilt, and the root `paper.pdf` is refreshed.

5.6 Milestones

5.7 Immediate next implementation

After this PDF tracker, the first concrete implementation target is **full active-practice coverage plus benchmark/correction scaffolding**. This is the first useful code milestone because transcription

Table 4. Milestone plan for moving from current evidence gates to solved long-phrase transcription.

Milestone	Target	Acceptance evidence	Status
M0	Paper tracker	Roadmap is included in the Curtis paper PDF and becomes the update record.	Complete in this revision.
M1	Practice time	All 213 h 23 min of strict-ledger video has active-practice coverage and per-day totals.	Coverage ledger/API and active-scan route added; live full scan pending.
M2	Score truth	Scherzo-Tarantelle has verified symbolic notes and score coordinates.	Pending.
M3	One-note anchor	A displayed A from audio is matched to a verified score-side A, with no wrong-note crop.	Pending.
M4	Short sequence	Three-to-five played notes align to exact score notes and render with local audio/video.	Pending.
M5	Long phrase	A 10–30 s passage transcription matches the paired audio and score location.	Pending.
M6	Repeat grouping	Multiple attempts of the same passage group into loops, restarts, slow attempts, and faster attempts.	Pending.
M7	Heat map	Practice density and problem density appear on verified score coordinates.	Pending.
M8	Repertoire update	Piece entries update from accepted daily evidence and show active time, clips, snippets, and blockers.	Pending.
M9	Readiness observation	Main blocker is generated from accepted evidence, not generic advice.	Pending.
M10	Regression lock	Tests prevent failed notation, wrong score boxes, and non-playing practice-time credit from returning.	Correction benchmark scaffold added; full benchmark suite pending.

cannot be trusted without knowing which source windows contain violin playing, and it cannot improve reliably without a gold correction loop.

The 2026-05-13 11:57 AM EDT backend update completed the coverage and correction scaffolding portion:

- added a canonical active-practice coverage ledger exposed through the review API and `/api/curtis/active-practice-coverage`;
- separated uploaded video duration, checked sampled video duration, measured active practice time, candidate violin windows, unmeasured video time, and estimate state;
- counted checked non-playing sampled windows as checked video without adding them to practice time;
- added `/api/curtis/evidence-corrections` for note, score, match, piece, and active-interval corrections;
- added benchmark fixtures for rejected score-note and match corrections;
- added regression tests for checked non-playing windows, wrong score-note benchmark creation, and rejecting accepted score evidence when the displayed score note conflicts with the observed note.

The 2026-05-13 12:33 PM EDT backend update completed the active-scan route portion:

- added a persistent `activePracticeScan` state record with active intervals, sample results, pending windows, and run history;
- scans stored local media samples only after the violin-presence gate is positive;
- extracts sound-active sub-intervals inside the source window and stores them as active-practice intervals;
- records negative violin-presence samples as checked without adding practice time;
- feeds persisted active intervals into per-day totals and the active-practice coverage ledger;
- exposes active-scan status and run endpoints so the deployed runtime can perform the pass without transcription;
- queues the next unchecked 90-second strict-ledger windows for owner media capture.

The remaining M1 work is running the deployed active-scan endpoint on the live media state, then continuing the low-cost media pass over every unmeasured strict-ledger video. The remaining M10 work is the larger benchmark suite for repeated notes, arpeggios, fast runs, score-free exercises, score-coordinate truth, and accepted notation/media completeness.

The next coding pass should:

1. run a low-cost active-practice pass over all remaining unchecked videos;
2. expose per-day uploaded duration, checked duration, detected active time, and estimate state;
3. persist exact active intervals produced by that pass;
4. expand the benchmark fixtures beyond the known wrong-score-note failure mode;
5. update this paper with the new coverage numbers and any solved milestone.

Only after M1 and the correction/benchmark loop are in place should the system try to claim longer transcription or score matching. Otherwise Curtis will keep producing plausible-looking fragments that Alan has to audit manually, which is the exact burden this roadmap is meant to remove.

6 Materials and methods

Backend

The backend is a FastAPI service deployed on Railway. It exposes `https://curtis.aolabs.io/api/curtis/media-status` for compact state, `https://curtis.aolabs.io/api/curtis/ops-check` for fuller operational state, local stored sample media for embedded playback after the media is violin-positive, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

Source scan

The default public source is `https://www.youtube.com/@nalalan`. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The media probe now keeps one early baseline window but then probes deeper quarter, midpoint, five-eighths, three-quarter, and tail windows before spending additional passes on early setup probes, and each stored sample is keyed by video identifier plus window start so later passes can expand coverage without overwriting earlier evidence. When configured above the dense-sampling threshold, the probe and owner-media helper generate dense 90-second windows from the start of each video through the latest valid window so the full practice archive can be checked without the earlier retention cap. Each sample is then classified by the violin-presence sampler before it can count as active practice evidence. The owner-media helper also checks its local cache of already captured browser samples, reuses classifier results when the file signature and classifier version match, prioritizes unsynced violin-positive cached windows before capturing more blind windows, and expands around already-found violin-positive windows before returning to blind sampling. The practice-hours total uses a stricter title-confirmed ledger from `violin 1` onward: violin-numbered videos and date-titled logs count toward the top-line session duration, while unrelated long videos remain indexed but outside the practice-hours total. The scan does not itself inspect performance quality.

Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, setup or non-playing windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes only for the source where the correction applies, so later videos can still identify that repertoire when present. A source correction also invalidates older confirmed labels for that same source instead of promoting the next stale candidate, and repeated corrected false labels keep that source out of the confirmed repertoire list until a stronger acceptance path exists. Rejected proposed titles, top candidates, and verification titles are scrubbed from current API output, not only from confirmed piece records. The 5/1 forensic pass rejected prior Paganini, Bruch, Mendelssohn, Brahms, Sibelius, Tchaikovsky, Wieniawski, Saint-Saens, Ravel, Bazzini, Ernst, Carmen, Zigeunerweisen, Bach Partitas, Kreisler Praeludium and Allegro, Sarasate family labels, Ysaye Sonata No. 3 Ballade, Mozart K. 216, Glinka Ruslan and Lyudmila, Strauss Till Eulenspiegel, Strauss Don Juan, Smetana Bartered Bride, Prokofiev Classical Symphony, Dvorak Carnival Overture, Shostakovich Symphony No. 5, Ravel Bolero, the first broad orchestral candidate batch, and the later overture, dance, and pops-orchestral candidate batch for that source. Alan then supplied the ground-truth source label: Haydn Symphony No. 94, IV. Finale, Violin I part. The system now treats that label as human-confirmed for 5/1, stops injecting new seeded guesses for the same source, and keeps Curtis-level completion unscored until judged playing evidence supports a score. Alan also supplied the 5/2 source label, Wieniawski Scherzo-Tarantelle, Op. 16, and then confirmed that all violin footage on 5/3 was the same piece. The training state now records all three source labels as confirmed sources and reports zero score-aligned windows until pitch/rhythm extraction has matched a source window to a score section. Transcribed samples can now produce learned pitch/rhythm fingerprints only after their media window is violin-positive, and visible notation requires note/rhythm verification. Future samples can receive a nearest-source hint when their extracted notes and rhythms resemble a prior confirmed source, an explicit title-labeled calibration upload, a symbolic reference, or a public labeled YouTube reference indexed through the official Data API; if the extracted pattern does not belong to a known score, it remains piece-or-exercise pending rather than being forced into repertoire. Public reference metadata can improve candidate vocabulary and target selection, but it does not become an audio fingerprint until a permitted media source exists. Possible labels and model-only verified titles remain uncertain daily evidence unless a source label is confirmed, so future misses appear as missing evidence rather than as wrong repertoire. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. The daily-record builder joins the strict `violin 1` ledger to violin-positive media samples, hidden machine note/rhythm events, practice-time evidence, clips, score packet support, repeated-fragment or repeated-pattern heat maps, score-aware read fields, and observation records. Repertoire entries are built only from confirmed daily-record evidence and keep progress percentages unassigned until clips, notation, and score or pattern alignment support them.

Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

7 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The broad micro-transcription gate is currently held closed: candidate note runs can be stored, but no displayed passage transcription should be treated as solved until its notes audibly match the paired clip. The only current displayed notation is two strict audio-checked note fragments, so the useful claim is limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

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